

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2015

Mathematics

Paper 2

Higher Level

Time: 2 hours, 30 minutes

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times, you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

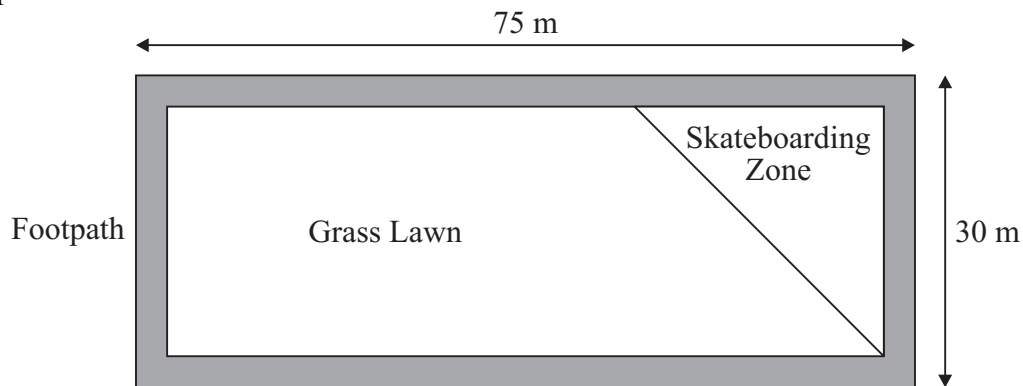
You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 20 minutes)

- (a)** The diagram below shows the plan for a rectangular park with sides measuring 75 m and 30 m. The park will consist of a grass lawn, a skateboarding zone with a dyed tarmacadam surface, in the shape of an isosceles triangle, and a 3 m-wide footpath around the edge of the park.



- (i) Find the area of the skateboarding zone.

[illegible]

- (ii) Find the area of the grass lawn.

[illegible]

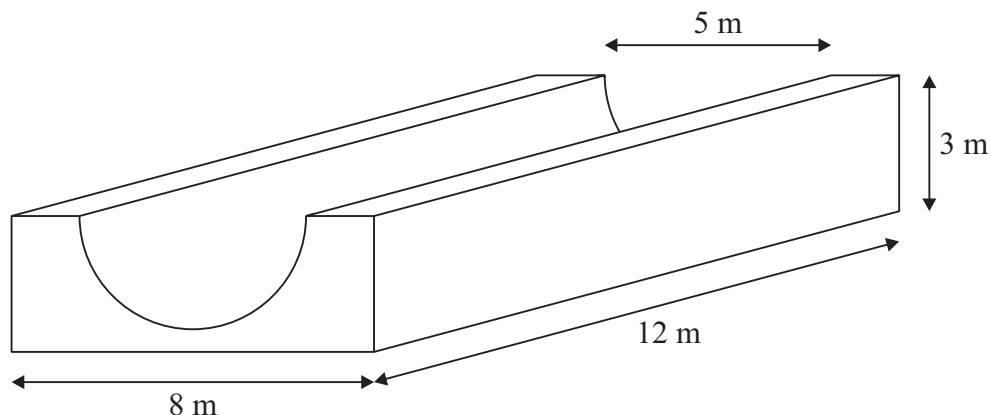
- (iii)** Find the area of the footpath around the edge of the park.

[illegible]

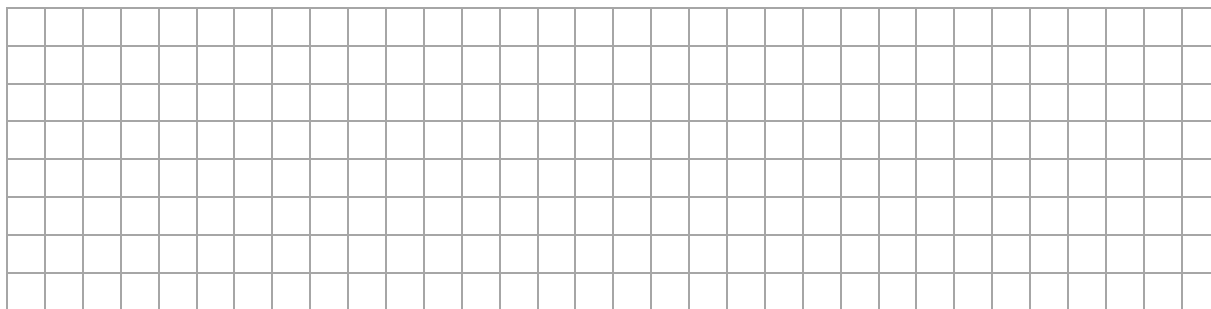
- (iv) A safety fence is to be installed around the perimeter of the skateboarding zone. Find the total length of fencing required. Give your answer in metres, correct to the nearest whole number.

[illegible]

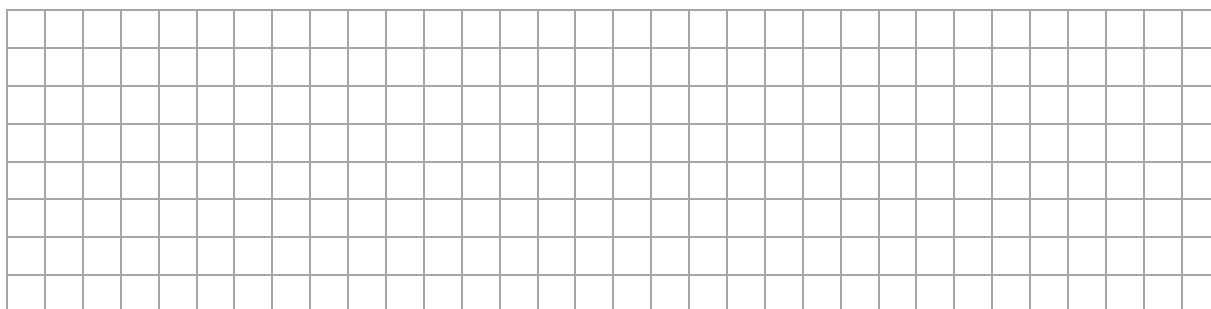
- (b) The skateboarding zone is to include a half-pipe made from concrete with a rubber membrane fitted to its curved surface as well as the two top landing areas either side of the curved surface. The membrane will protect the surface of the half-pipe against weathering and wearing due to skateboards. The half-pipe comprises of a rectangular cuboid with a semi-cylinder removed, as shown. The dimensions are shown in the diagram below.



- (i) Find the volume of concrete used in the construction of the half-pipe.
Give your answer correct to one decimal place.

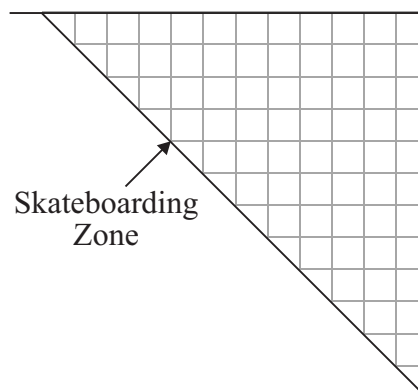


- (ii) Find the total surface area that is to be covered by the rubber membrane.
Give your answer correct to one decimal place.



- (iii) A scaled diagram of the skateboarding zone is shown on the grid below. Each grid box represents a $2\text{ m} \times 2\text{ m}$ square.

On the diagram, indicate a suitable location for the half-pipe.



- (c) A building company wishes to put in a bid for the construction of the park. The company's costs for the different elements of the project (inclusive of labour) are shown in the table below.

Item	Cost
Dyed tarmacadam for the skateboarding zone	€17 per m ²
Green lawn (including planting of trees and shrubs)	€110 per m ²
Tarmacadam for footpath	€15 per m ²
Concrete for half-pipe	€45 per m ³
Rubber membrane for half-pipe	€22 per m ²
Fencing for perimeter of the skateboarding zone	€14 per m

- (i) Using your answers from parts (a) and (b), calculate the company's total cost to construct the park.

A full-page sheet of white graph paper featuring a uniform grid of thin, light gray horizontal and vertical lines. The grid consists of small squares covering the entire area of the page.

- (ii)** The building company put in a successful bid of €210 000 for the construction of the park. Find the percentage profit that the company will make if it completes the construction for the cost calculated in part **(c)(i)**. Give your answer correct to one decimal place.

Question 2

(Suggested maximum time: 15 minutes)

Claire carried out a survey in her school and then displayed the results in the two-way table below.



	Right-handed	Left-handed
Girls	186	34
Boys	203	37

- (a) Based on the above data, write the **two** questions Claire asked in the survey in a suitable form.

[illegible]

- (b)** What type of data did Claire collect in the survey? Put a tick (✓) in the correct box below. Explain your answer.

Numerical Discrete

7

Numerical Continuous

1

Categorical
Nominal

1

Categorical Ordinal

7

[illegible]

- (c) What percentage of students surveyed are left-handed? Give your answer correct to one decimal place.

[illegible]

- (d) If one student is chosen at random from the survey, what is the probability that the student chosen is not a right-handed girl?

[illegible]

- (e) Claire says: “It is more likely to find a left-handed boy than a left-handed girl in her school.” Based on the data collected, do you agree with Claire? Give a reason for your answer.

Answer:

Reason:

- (f) If one boy and one girl are chosen at random from the survey, what is the probability that both students chosen are left-handed?

[illegible]

Question 3

(Suggested maximum time: 15 minutes)

The prices of twelve pairs of designer shoes to buy in-store are shown in the incomplete back-to-back stem-and-leaf plot below.



In-store						Online					
				9	3						
			5	4	4						
	7	5	0	0	5						
		9	6	5	6						
				5	7						
				3	8						
								Key: 6 5 = €65			

- (a) (i)** Complete the stem-and-leaf plot to display the online prices of the same shoes shown in the table below.

Online Prices (€)		
43	59	31
56	76	42
67	48	54
46	58	63

- (ii)** Use your diagram to identify whether shoes are cheaper to buy in-store or online. Explain your answer.

[illegible]

- (b) (i)** What is the in-store modal price?

[illegible]

- (ii) Aoife says that she cannot find the online modal price. Explain why this is the case.

[illegible]

- (c) Find the median price in each case.

In-store:	Online:

- (d)** Where does the price of designer shoes have the largest range, in-store or online? Give a reason for your answer.

[illegible]

- (e) Find the interquartile range in each case.

[illegible]

- (f) Compare the prices of designer shoes to buy in-store and online. Refer to **at least one** measure of central tendency and **at least one** measure of variability (spread) in your answer.

[illegible]

- (g) Laura says: “Every pair of designer shoes is cheaper online.” Do you agree with her? Give a reason for your answer.

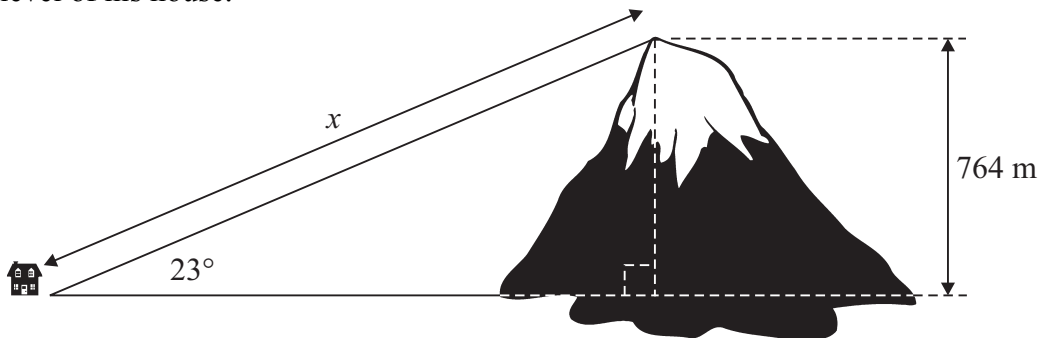
[illegible]

Question 4

(Suggested maximum time: 10 minutes)

- (a) Mark wishes to find x , the distance ‘as the crow flies’ from his home to the top of a nearby mountain. He uses a clinometer to measure the angle of elevation to the top of the mountain and records an angle of 23° .

He then looks at a map of his local area and finds that the top of the mountain is 764 m above the ground level of his house.



- (i) Use these measurements to calculate x , correct to one decimal place.

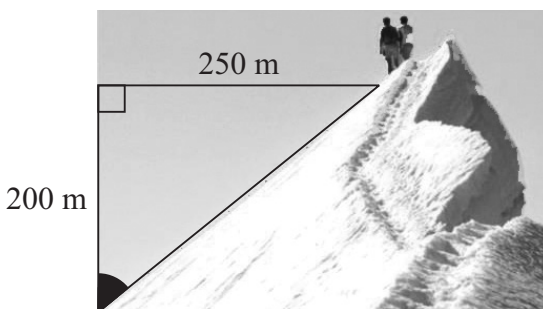
[illegible]

- (ii) Suggest **one** possible source of error that may affect Mark's calculation.

[illegible]

- (b)** To prevent dangerous avalanches near ski resorts in Switzerland, the Swiss authorities initiate sets of mini-avalanches when the gradient of snowfall on the top of mountains reaches 45° to the vertical.

- (i) Calculate the angle between the gradient of the snowfall and the vertical, as shown in the diagram. Give your answer in degrees and minutes (correct to the nearest minute).

[illegible]

Question 6

(Suggested maximum time: 10 minutes)

Aer Lingus operates scheduled flights between five airports in Ireland and seventeen airports in Britain.



- (a)** How many different possible flight routes can Aer Lingus operate between Ireland and Britain?

[illegible]

- (b)** The actual number of scheduled flights operated by the airline on an average weekday between Ireland and Britain is 210. Suggest **two** reasons why this number is different to your answer in part **(i)**.

[illegible]

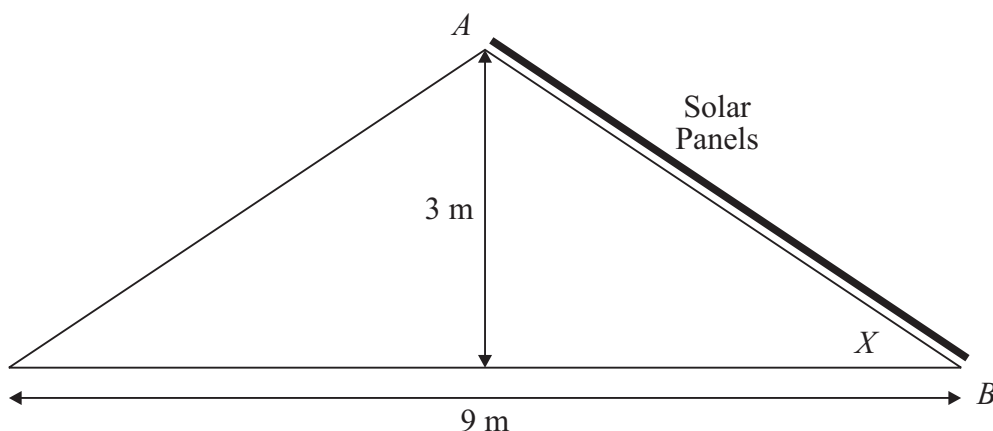
- (c) On a particular Monday, Aer Lingus only operated 165 flights between the two countries due to bad weather in Britain. Given that flights between Ireland and Britain represent two-thirds of the airline's daily scheduled flights, calculate the overall percentage decrease in the number of flights operated by the airline on that day. Give your answer correct to one decimal place.

A large grid of graph paper with 20 columns and 15 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

Question 7

(Suggested maximum time: 10 minutes)

The installation of solar panels on the roof of a house can reduce the energy bills of the household. The cross section of the roof of a house which is being fitted with solar panels is shown below. The total span of the roof is 9 m and its height is 3 m. The apex of the roof is directly above the midpoint of the roof span.



- (a) In which direction should the solar panels face on the roof of the house to maximise the impact of the sun? Give a reason for your answer.

[illegible]

- (b)** Identify the type of triangle which represents the cross section of the roof shown in the diagram above. Give a reason for your answer.

[illegible]

- (c) Calculate the measure of the angle X , correct to one decimal place.

[illegible]

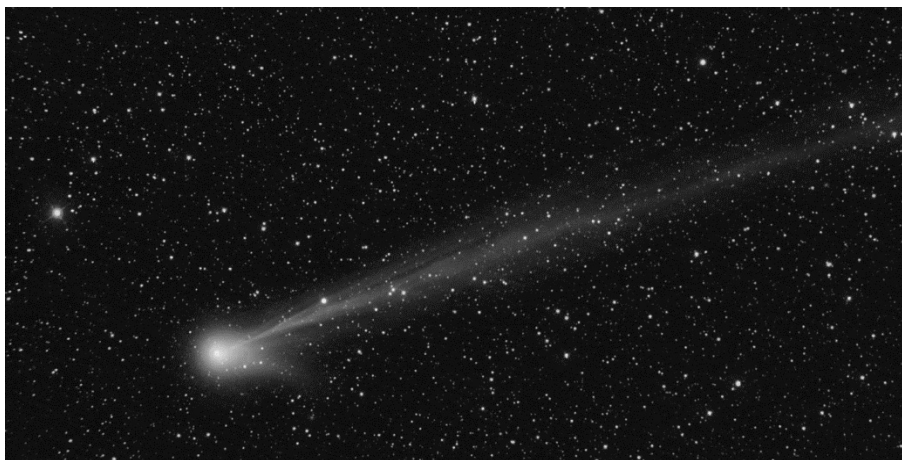
- (d) Hence, find $|AB|$, the maximum length of solar panels that can be fitted to the roof. Give your answer correct to two decimal places.

[illegible]

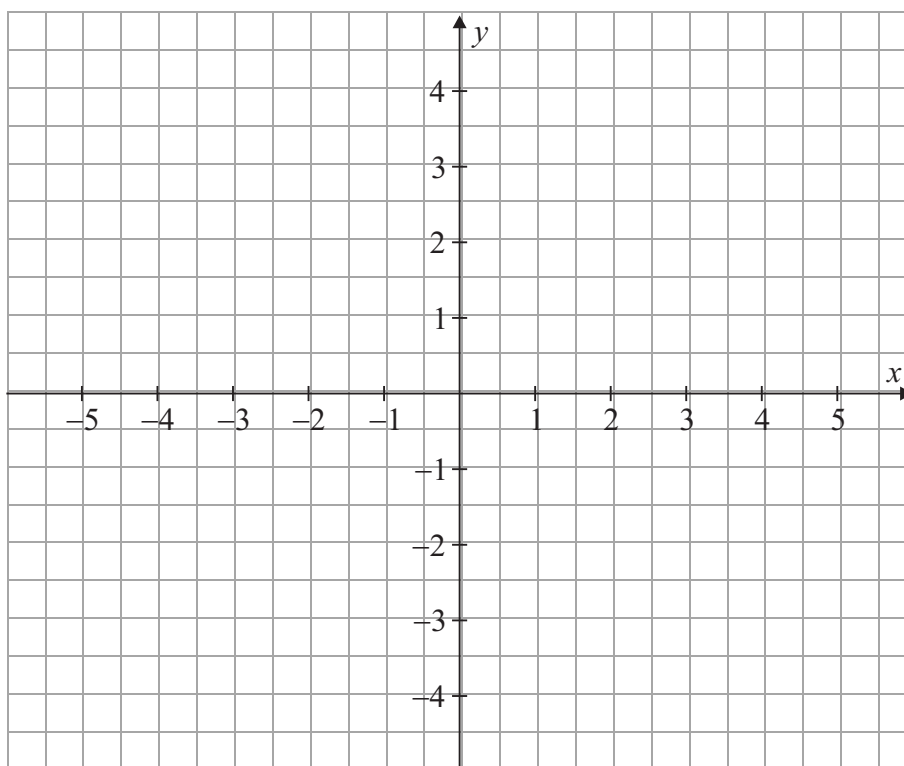
Question 8

(Suggested maximum time: 20 minutes)

Scientists at the European Space Agency (ESA) monitor the flight paths of comets and asteroids that may collide with Earth. The progress of one particular asteroid is tracked over a 24-hour period. A researcher decided to plot the course of the asteroid on a co-ordinate grid to see how close it came to a collision. At the start of the tracking period, the asteroid was located at point $A(-3, 2)$ and at the end of the tracking period, it was located at point $B(4, -3)$. Earth is located at the point $(0, 0)$.



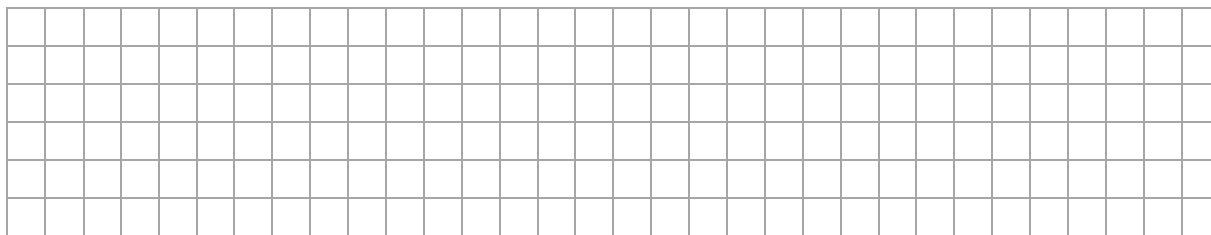
- (a) Plot the points A and B on the co-ordinate plane below. Hence, draw the line that represents the flight path of the asteroid.



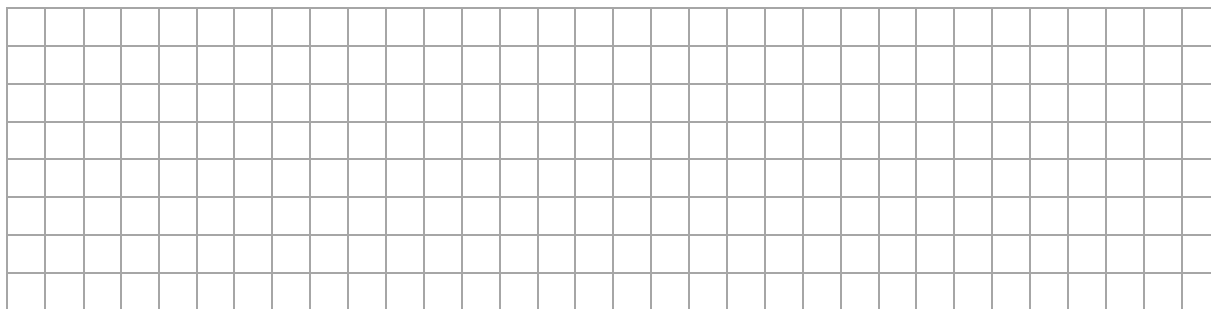
- (b)** Find the slope of AB .

[illegible]

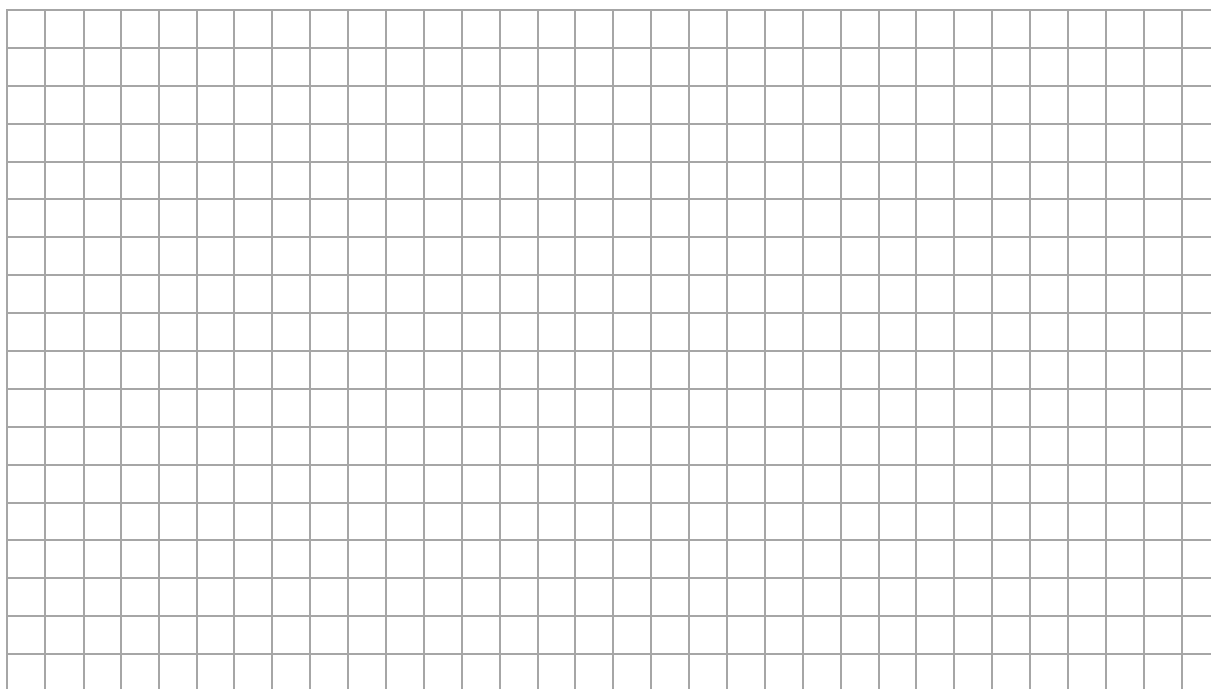
- (c) Find the equation of the line AB .



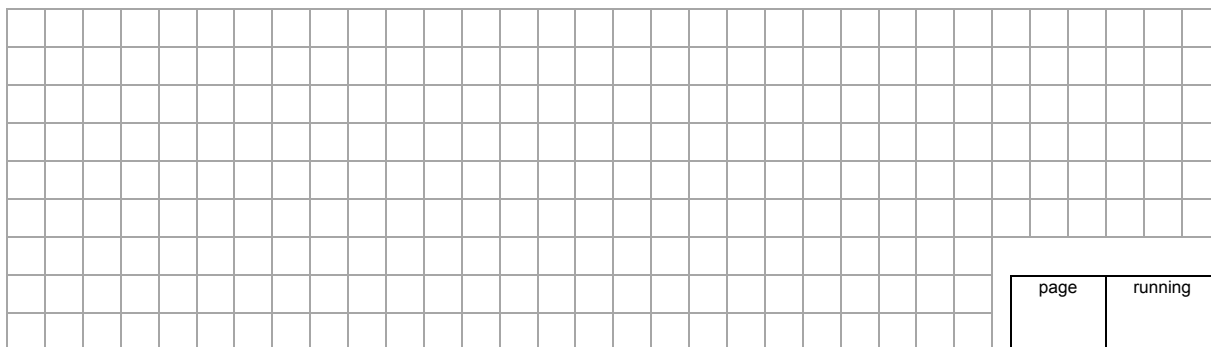
- (d) Find the equation of the line through $(0, 0)$, perpendicular to AB .



- (e) Let C be the point where this perpendicular line through $(0, 0)$ intersects AB . Calculate the co-ordinates of C .



- (f) The co-ordinate grid is drawn to a scale of 1 unit = 500 000 km. How close did the asteroid come to a collision with Earth? Give your answer correct to the nearest kilometre.



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Question 9

(Suggested maximum time: 5 minutes)

- (a)** Examine the letters of the word shown below.

MATHEMATICS

- (i) Which letter has no axes of symmetry?

[illegible]

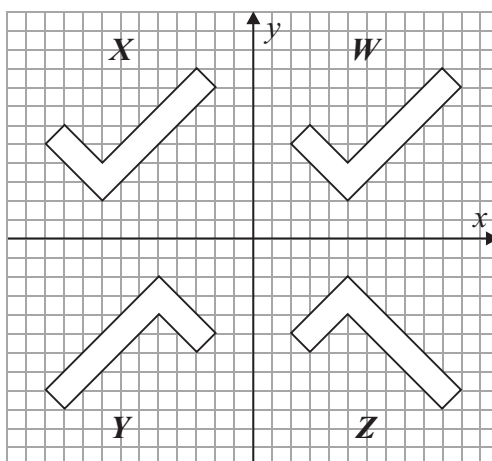
- (ii)** Identify **one** letter which has only one axis of symmetry.

[illegible]

- (iii) Which **two** letters which have more than one axis of symmetry?

[illegible]

- (b)** Each of the three images X , Y and Z in the diagram below shows the image of the object W under a transformation.



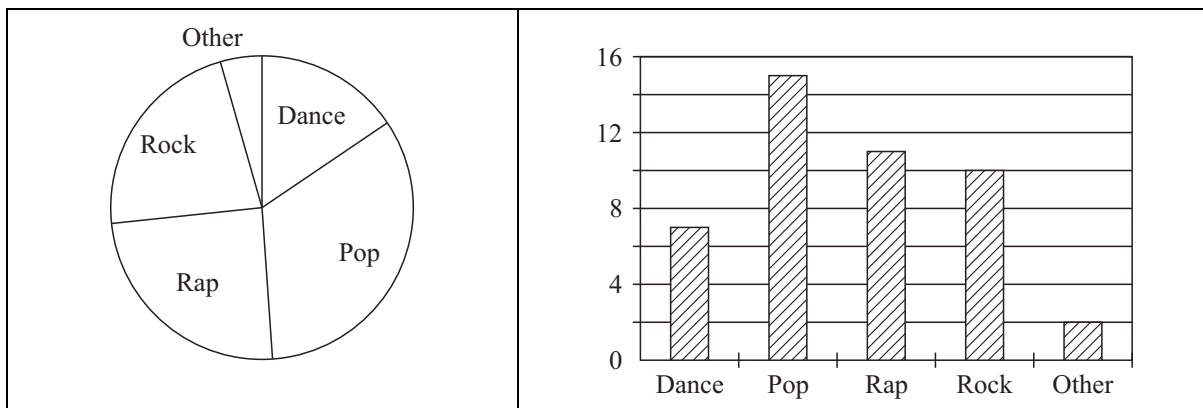
For each of X , Y and Z , describe fully one transformation (translation, axial symmetry or central symmetry) that will map the object onto that image.

[illegible][illegible][illegible]

Question 10

(Suggested maximum time: 10 minutes)

Beth has been asked to organise the music selection for an upcoming disco for students at her school. She conducted a survey by asking a group of girls in her year group to indicate their preferred choice of music. Her results are shown in the two charts below.



- (a) (i)** Beth wants to show that pop music was the most popular choice. Which chart do you think she should use? Give a reason for your answer.

[illegible]

- (ii)** Identify another suitable type of graph that Beth could use to display this data.

[illegible]

- (b)** Calculate the measure of the angle in the above pie chart that represents the students who indicated rap music as their preferred choice of music in the survey.

[illegible]

- (c) Beth claims that the results of her survey are representative of the views of the mixed student population in her school. Do you agree with Beth? Give a reason for your answer.

Answer:	
Reason:	
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Question 11

(Suggested maximum time: 15 minutes)

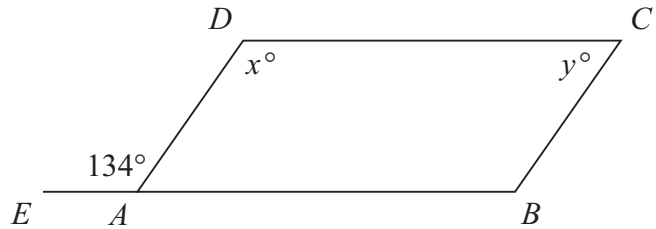
- (a)** Prove that in a parallelogram, opposite sides are equal and opposite angles are equal.

Diagram:

<i>Given:</i>	
<i>To Prove:</i>	
<i>Construction:</i>	
<i>Proof:</i>	

- (b) $ABCD$ is a parallelogram, as shown.
 BA has been extended to E .
 $|\angle EAD| = 134^\circ$, as shown.

Find the measure of the angle x°
 and the measure of the angle y° .



$x^\circ =$										$y^\circ =$									

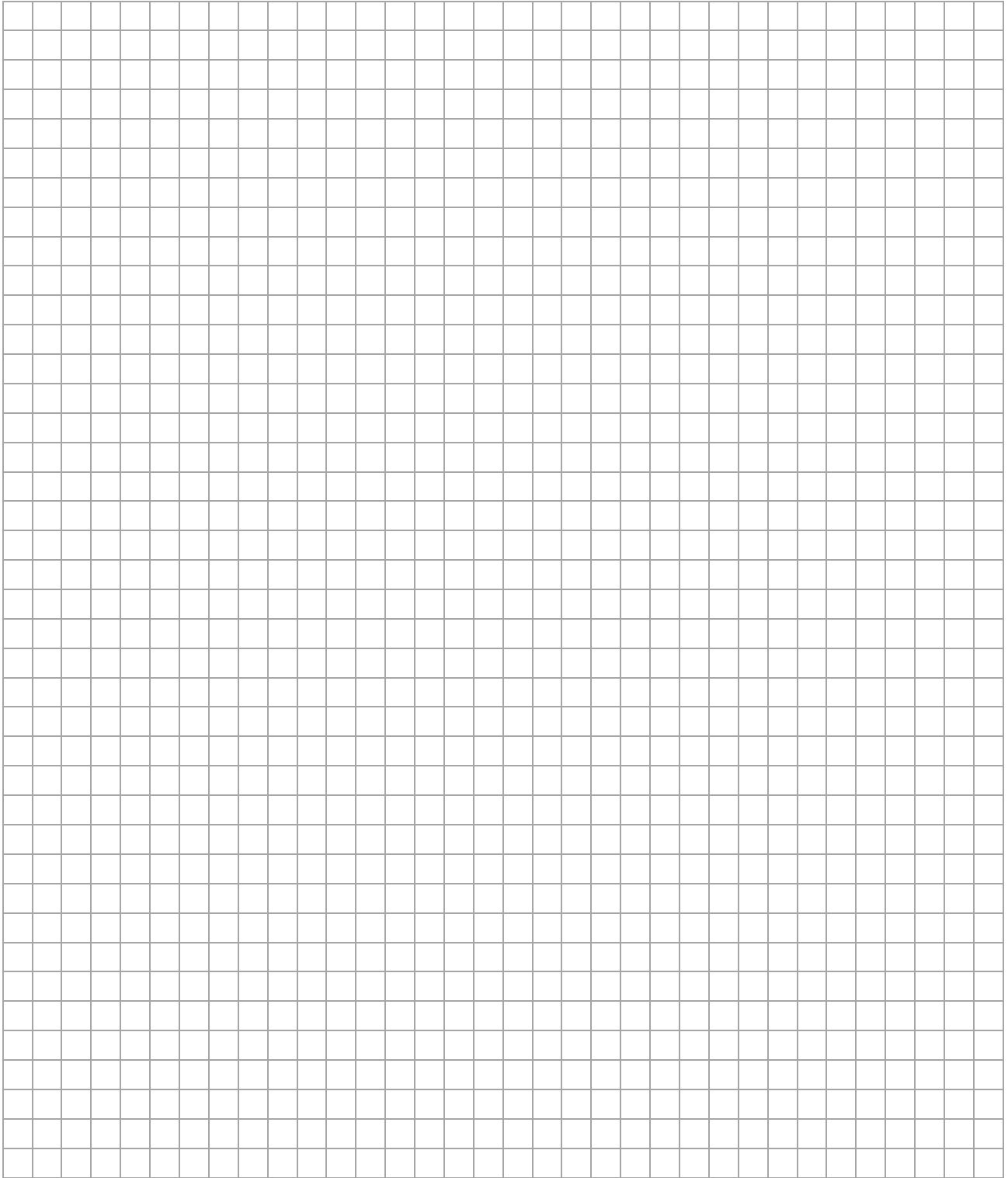
Question 12

(Suggested maximum time: 5 minutes)

Using only a compass and a straight edge, construct a line through the point A perpendicular to the line l below. Show all construction work.

• A

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Pre-Junior Certificate, 2015 – Higher Level

Mathematics – Paper 2

Time: 2 hours, 30 minutes

